Key concepts

TRANSFORMATION OF INDEPENDENT VARIABLE

Time shift: Two signals identical in shape, but that are displaced or shifted relative to each other.

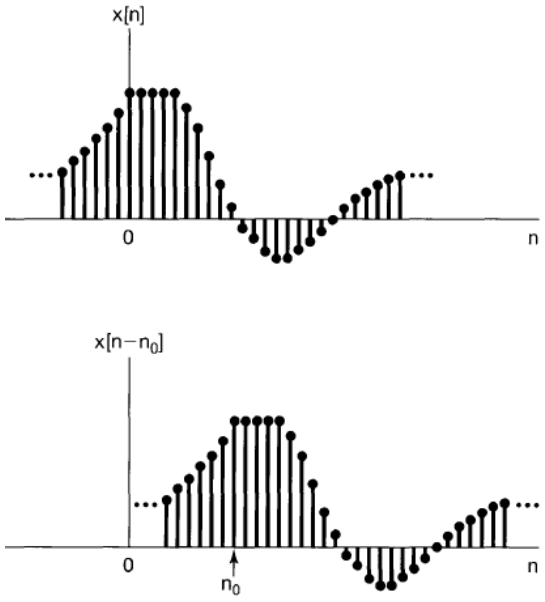


Figure 1.8 Discrete-time signals related by a time shift. In this figure $n_0 > 0$, so that $x[n - n_0]$ is a delayed verson of $x[n]$ (i.e., each point in $x[n]$ occurs later in $x[n - n_0]$).

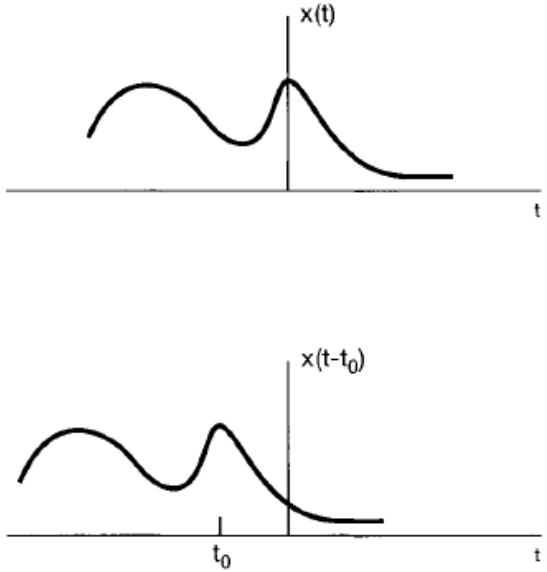


Figure 1.9 Continuous-time signals related by a time shift. In this figure $t_0 < 0$, so that $x(t - t_0)$ is an advanced version of $x(t)$ (i.e., each point in $x(t)$ occurs at an earlier time in $x(t - t_0)$).

TRANSFORMATION OF INDEPENDENT VARIABLE

Time reversal: the signal obtained by a reflection about n = 0 (i.e., by reversing the signal).

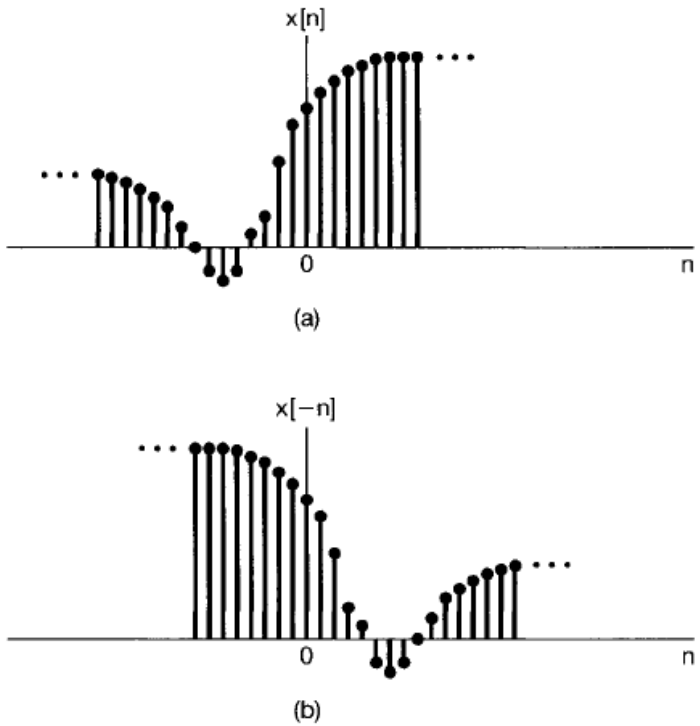


Figure 1.10 (a) A discrete-time signal $x[n]$; (b) its reflection $x[-n]$ about $n = 0$.

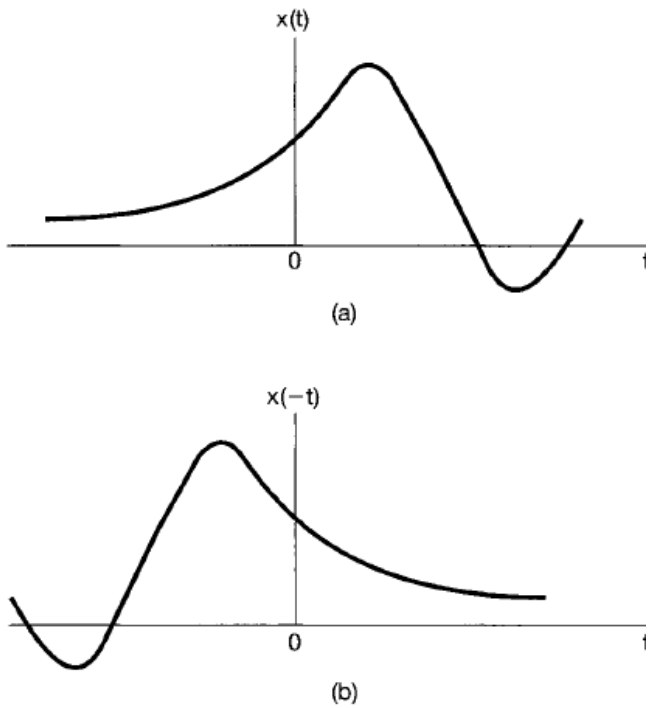


Figure 1.11 (a) A continuous-time signal $x(t)$; (b) its reflection $x(-t)$ about $t = 0$.

TRANSFORMATION OF INDEPENDENT VARIABLE

Time scaling: Signals related by scale changes in the independent variable.

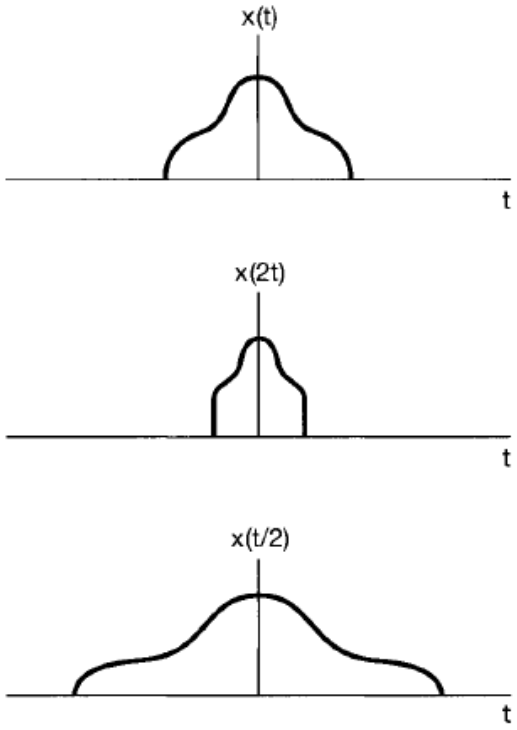
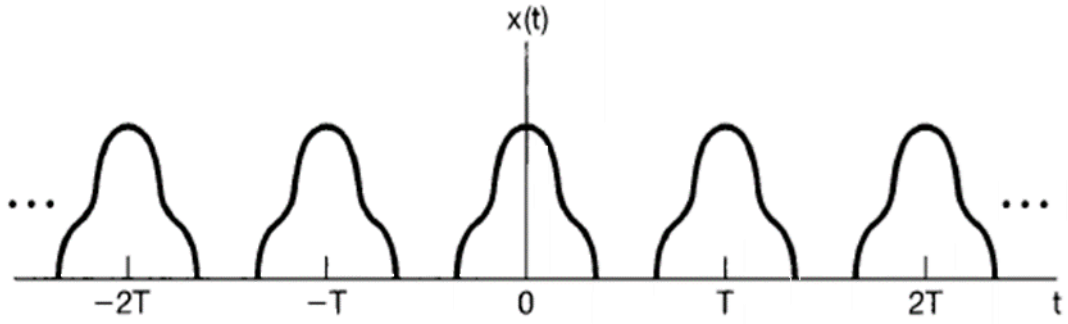


Figure 1.12 Continuous-time signals related by time scaling.

PERIODIC C.T. SIGNAL

A signal that repeats itself after a certain time shift T_0

$x(t) = x(t + T_0)$ period :=smallest T_0



$x(t)$ repeats for $x(t - T)$, $x(t + T)$, $x(t - 2T)$, $x(t + 2T)$ ...

PERIODIC C.T. SIGNAL

$$x(t) = x(t + T_0)$$

period :=smallest T_0

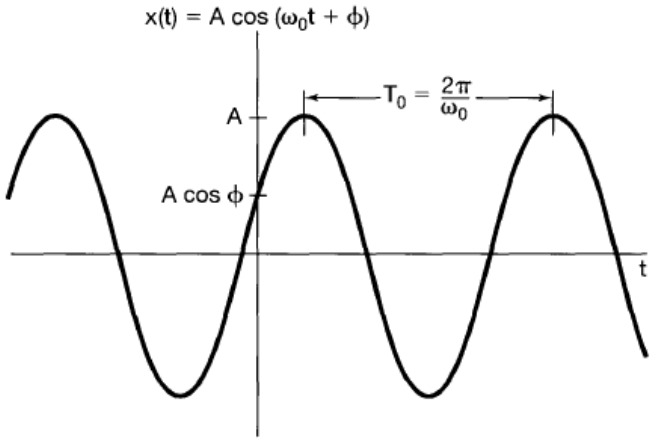


Figure 1.20 Continuous-time sinusoidal signal.

For the sinusoidal signal:

$$A\cos[w_0 t + \phi] = A[w_0(t + T_0) + \phi] = A\cos[w_0 t + \underbrace{w_0 T_0}_{2\pi m} + \phi]$$

For the left hand side to be equal to right hand side:

$$w_0 T_0 = 2\pi m$$

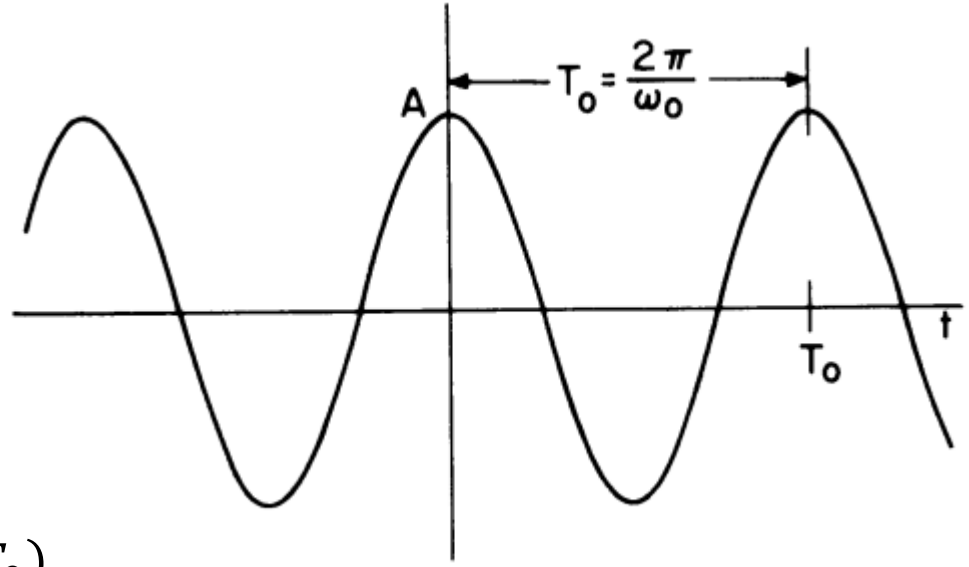
$$T_0 = \frac{2\pi m}{w_0}$$

For the smallest T_0

$$\textbf{Period} = \frac{2\pi}{w_0}$$

EVEN C.T. SIGNALS

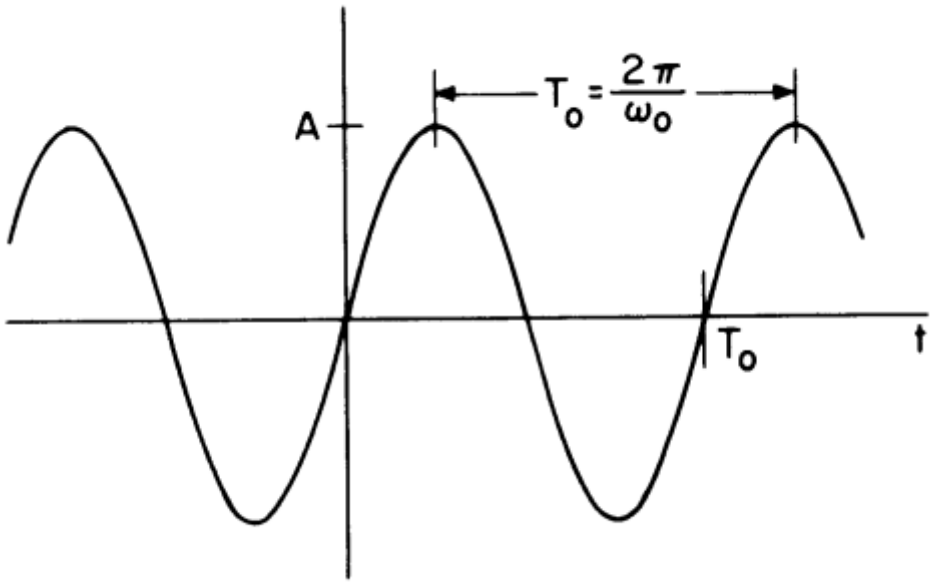
$x(t) = Acos(w_0 t + \phi)$ where ϕ=0



- Periodic: $x(t) = x(t + T_0)$
- Even: $x(t) = x(-t)$

ODD C.T. SIGNALS

$$x(t) = \begin{cases} A\cos(w_0 t - \frac{\pi}{2}) \\ A\sin(w_0 t) \\ A\cos\left[w_0\left(t - \frac{T_0}{4}\right)\right] \end{cases} ; \text{so } \phi = -\frac{\pi}{2}$$



- Periodic: $x(t) = x(t + T_0)$
- Odd: $x(t) = -x(-t)$

PERIODIC D.T. SIGNALS

$$x[n] = A\cos(\Omega_0 n + \phi)$$

Periodic: $x[n] = x[n+N]$, smallest integer N := period

PERIODIC D.T. SIGNALS

For the sinusoidal signal:

$$x[n] = A\cos(\Omega_0 n + \phi)$$

Periodic: $x[n] = x[n+N]$, smallest integer N := period

$$x[n] = A\cos(\Omega_0(n+N) + \phi)$$

$$= A\cos(\Omega_0 n + \underbrace{\Omega_0 N}_{2\pi m} + \phi)$$

Periodic → $\Omega_0 N = 2\pi m$

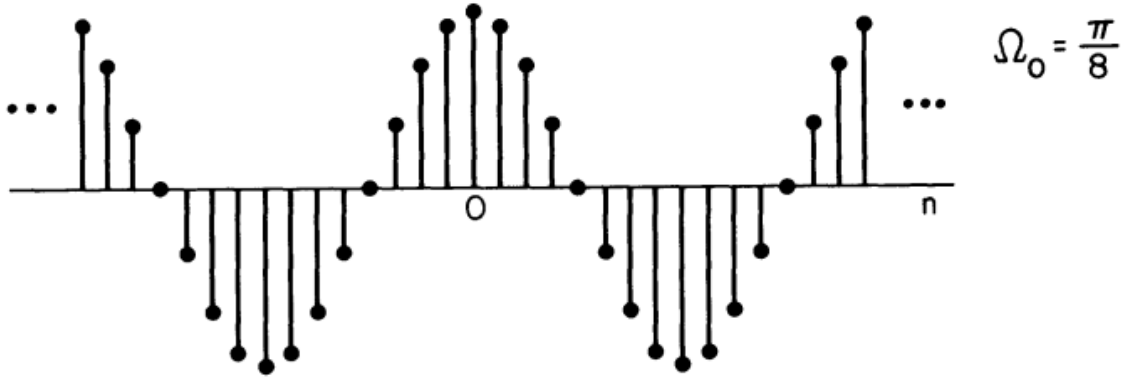
$$\boldsymbol{N = \frac{2\pi m}{\Omega_0}}$$

N, m must be integers

Smallest N (if any) = period

EVEN D.T. SIGNALS

$x[n] = Acos(\Omega_0 n)$, ϕ=0

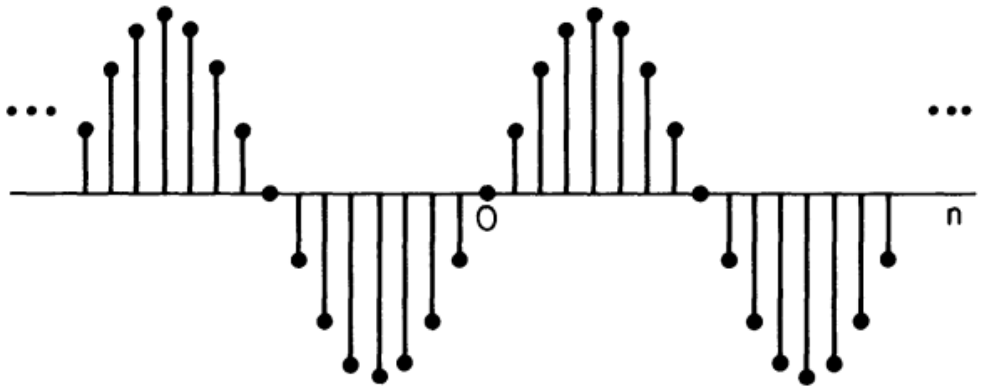


- Even: $x[n] = x[-n]$

ODD D.T. SIGNALS

$$x[n] = \begin{cases} A\cos(\Omega_0 n - \frac{\pi}{2}) \\ A\sin(\Omega_0 n) \\ A\cos[\Omega_0(n - n_0)] \end{cases} ; \text{so } \phi = -\frac{\pi}{2}$$

n_0=?



$\Omega_0 = \frac{\pi}{8}$

- Odd: $x[n] = -x[-n]$

EVEN AND ODD SIGNAL DECOMPOSITION

$$\mathcal{E}v\{x(t)\} = \frac{1}{2}[x(t) + x(-t)]$$

$$\mathcal{O}d\{x(t)\} = \frac{1}{2}[x(t) - x(-t)]$$

$$\mathcal{E}v\{x[n]\} = \frac{1}{2}\big[x[n] + x[-n]\big]$$

$$\mathcal{O}d\{x[n]\} = \frac{1}{2}\big[x[n] - x[-n]\big]$$

Same for C.T. and D.T.!

Example: $x[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$

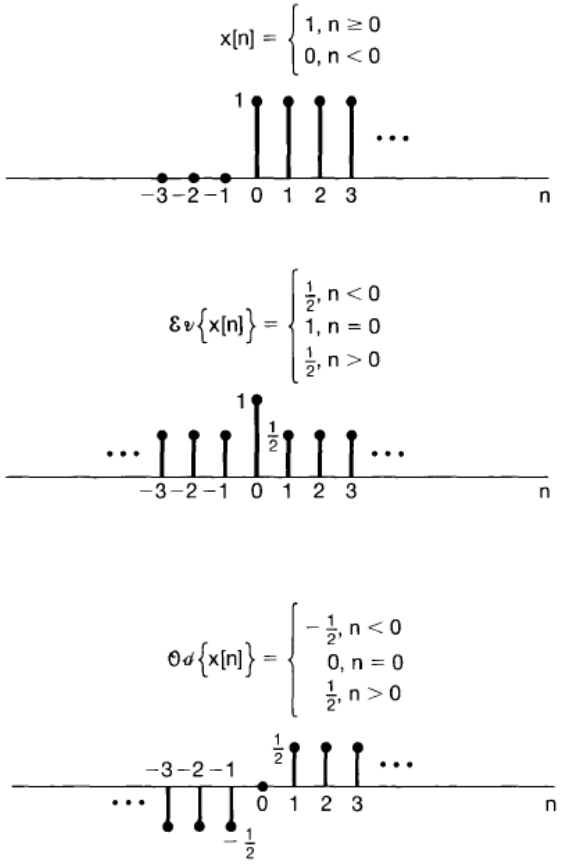


Figure 1.18 Example of the even-odd decomposition of a discrete-time signal.

SYSTEMS WITH AND WITHOUT MEMORY

A system is said to be *memory less* if its output for each value of the independent variable at a given time is dependent only on the input at that same time.

$$y(t) \text{ @ } t = t_0 \leftarrow x(t) \text{ @ } t = t_0$$

$$y[n] \text{ @ } n = n_0 \leftarrow x[n] \text{@ } n = n_0$$

Examples:

$$y(t) = x^2(t)$$
$$y[n] = x^2[n]$$ ✓

$$y(t) = \int_{-\infty}^{t} x^2(\tau)d\tau$$
$$y[n] = \sum_{-\infty}^{n} x^2[n]$$ ✗

$$y(t) = Rx(t)$$
$$y[n] = Rx[n]$$ ✓

$$y[n] = x[n-1]$$ ✗

INVERTIBILITY AND INVERSE SYSTEMS

A system is said to be *invertible* if distinct inputs lead to distinct outputs.

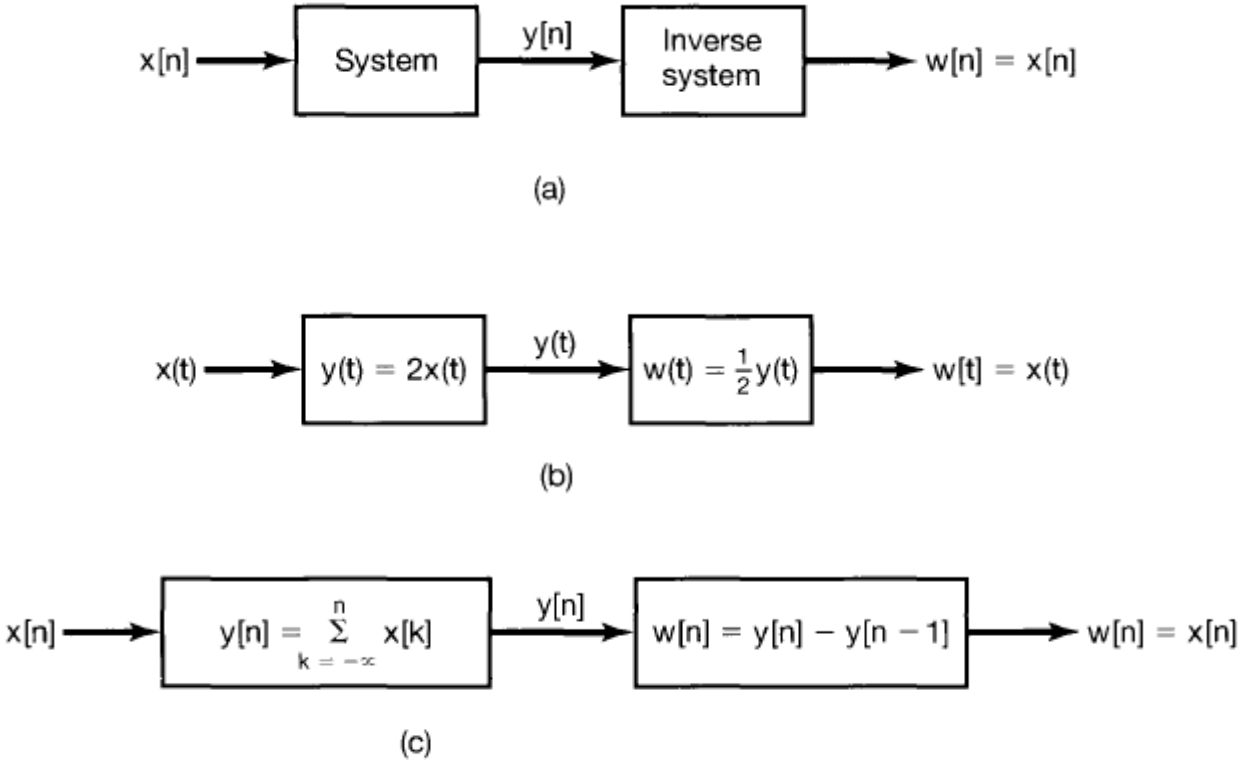


Figure 1.45 Concept of an inverse system for: (a) a general invertible system; (b) the invertible system described by eq. (1.97); (c) the invertible system defined in eq. (1.92).

$$y(t) = x^2(t)$$

Invertible? ✗

Memoryless? ✓

CAUSALITY

A system is said to be causal if the output at any time depends only on values of the input at the present time and in the past.

OR

The system can't anticipate future inputs

$$x_1(t) \rightarrow y_1(t)$$
$$x_2(t) \rightarrow y_2(t)$$

If

$$x_1(t) = x_2(t)\ for\ t < t_0$$

then

$$y_1(t) = y_2(t)\ for\ t < t_0$$

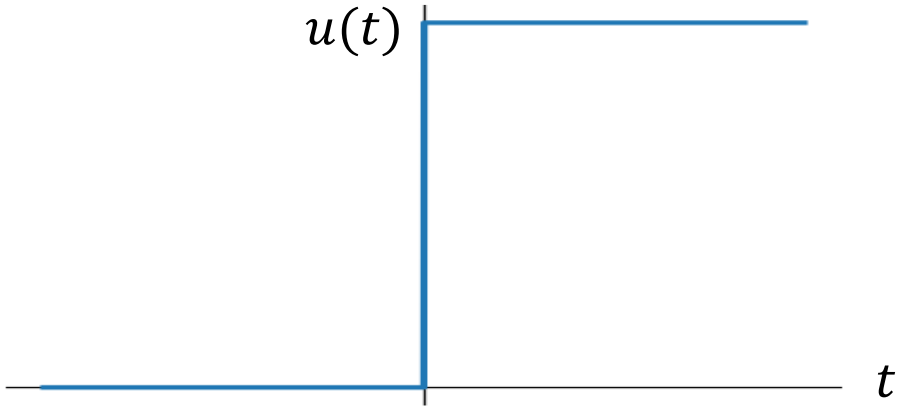
Same for discrete time.

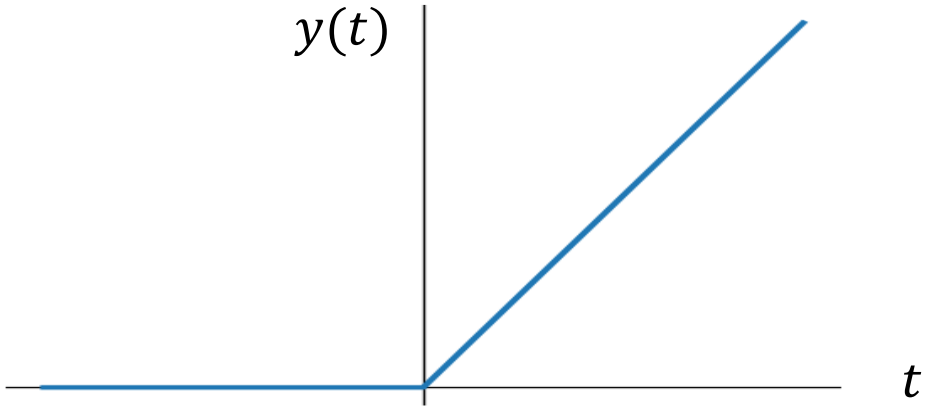
STABILITY

Bounded input, bounded output!

A system is stable when small inputs lead to responses that do not diverge.

$$y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$$





TIME INVARIANCE

A system is time-invariant if delaying the input to the system simply delays the output by the same amount of time.

C.T:

$x(t) \rightarrow y(t)$

Then

$x(t - t_0) \rightarrow y(t - t_0)$

D.T.:

$x[n] \rightarrow y[n]$

Then

$x[n - n_0] \rightarrow y[n - n_0]$

Example 1.14

Consider the continuous-time system defined by

$$y(t) = \sin[x(t)]. \tag{1.114}$$

To check that this system is time invariant, we must determine whether the time-invariance property holds for *any* input and *any* time shift t_0. Thus, let $x_1(t)$ be an arbitrary input to this system, and let

$$y_1(t) = \sin[x_1(t)] \tag{1.115}$$

be the corresponding output. Then consider a second input obtained by shifting $x_1(t)$ in time:

$$x_2(t) = x_1(t - t_0). \tag{1.116}$$

The output corresponding to this input is

$$y_2(t) = \sin[x_2(t)] = \sin[x_1(t - t_0)]. \tag{1.117}$$

Similarly, from eq. (1.115),

$$y_1(t - t_0) = \sin[x_1(t - t_0)]. \tag{1.118}$$

Comparing eqs. (1.117) and (1.118), we see that $y_2(t) = y_1(t - t_0)$, and therefore, this system is time invariant.

$$y[n] = nx[n]$$

Is this time invariant?

LINEARITY

A system is linear if its response to a weighted sum of inputs is equal to the weighted sum of its responses to each of the inputs.

C.T & D.T.

$x_1(t) \rightarrow y_1(t)$

$x_2(t) \rightarrow y_2(t)$

Then

$\alpha x_1(t) + \beta x_2(t) \rightarrow \alpha y_1(t) + \beta y_2(t)$

Example 1.17

Consider a system S whose input $x(t)$ and output $y(t)$ are related by

$$y(t) = tx(t)$$

To determine whether or not S is linear, we consider two arbitrary inputs $x_1(t)$ and $x_2(t)$.

$$x_1(t) \rightarrow y_1(t) = tx_1(t)$$
$$x_2(t) \rightarrow y_2(t) = tx_2(t)$$

Let $x_3(t)$ be a linear combination of $x_1(t)$ and $x_2(t)$. That is,

$$x_3(t) = ax_1(t) + bx_2(t)$$

where a and b are arbitrary scalars. If $x_3(t)$ is the input to S, then the corresponding output may be expressed as

$$\begin{aligned} y_3(t) &= tx_3(t) \\ &= t(ax_1(t) + bx_2(t)) \\ &= atx_1(t) + btx_2(t) \\ &= ay_1(t) + by_2(t) \end{aligned}$$

We conclude that the system S is linear.

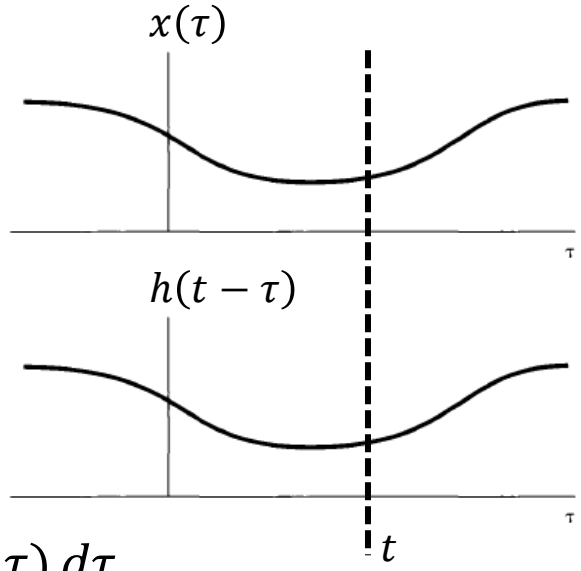
CONVOLUTION

DT: $$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

CT: $$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\, d\tau$$

PROPERTIES OF LTI SYSTEMS

Memory or Memoryless:



$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\, d\tau$$

Memoryless if $h(t-\tau)$ is non-zero only at $\tau = t$

$\Rightarrow h(t) = k\delta(t)$ in C.T

$h[n] = k\delta[n]$ in D.T

$\Rightarrow y(t) = kx(t)$

PROPERTIES OF LTI SYSTEMS

Invertibility:

$$w(t) = \big(x(t) * h(t)\big) * h_1(t) = x(t)$$

Apply associative property:

- $\mathrm{w}(t) = x(t) * \big(h(t) * h_1(t)\big)$

For $w(t) = x(t)$

- $\mathrm{w}(t) = x(t) * \underbrace{\big(h(t) * h_1(t)\big)}_{\delta(t)}$

- $h_1(t)$ is the inverse of $h(t)$
- $h_1(t)$=$h^{-1}(t)$

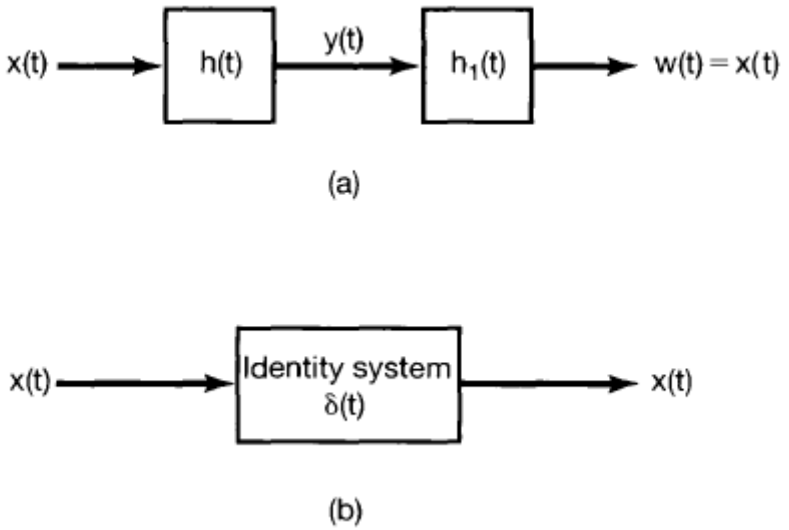


Figure 2.26 Concept of an inverse system for continuous-time LTI systems. The system with impulse response $h_1(t)$ is the inverse of the system with impulse response $h(t)$ if $h(t) * h_1(t) = \delta(t)$.

PROPERTIES OF LTI SYSTEMS

Stability:

- Bounded input, bounded output!

A system is stable if:

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

$$\int_{-\infty}^{\infty} |h(\tau)|\, d\tau < \infty$$

PROPERTIES OF LTI SYSTEMS

Causality:

$$x_1(t) \rightarrow y_1(t)$$
$$x_2(t) \rightarrow y_2(t)$$

If

$$x_1(t) = x_2(t)\ for\ t < t_0$$

then

$$y_1(t) = y_2(t)\ for\ t < t_0$$

For linear systems:

If

$$x(t) = 0\ for\ t < t_0$$

then

$$y(t) = 0\ for\ t < t_0$$

Zero in, zero out!

C.T. FOURIER SERIES

Determining harmonic components of a periodic signal.

Synthesis Equation:

$$x(t) = X(t + T_0) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

Analysis Equation:

$$a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt = \frac{1}{T}\int_T x(t)e^{-jk(\frac{2\pi}{T_0})\,t}dt$$

C.T. FOURIER TRANSFORM PAIR

One of the most useful features of the Fourier transform is the simple “inverse” Fourier transform.

“Synthesis” equation:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t} d\omega$$

“Analysis” equation :

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

C.T. FOURIER TRANSFORM OF PERIODIC SIGNALS

Fourier transform representations for periodic signals allow us to consider both periodic and aperiodic signals within a unified context.

$$X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0)$$

where

$$a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt = \frac{1}{T}\int_T x(t)e^{-jk(\frac{2\pi}{T_0})\,t}dt$$

PROPERTIES OF C.T. FOURIER TRANSFORM [CHECK YOURSELF]

TABLE 4.1 PROPERTIES OF THE FOURIER TRANSFORM

Section	Property	Aperiodic signal	Fourier transform
		$x(t)$ $y(t)$	$X(j\omega)$ $Y(j\omega)$
4.3.1	Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
4.3.2	Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0}X(j\omega)$
4.3.6	Frequency Shifting	$e^{j\omega_0 t}x(t)$	$X(j(\omega - \omega_0))$
4.3.3	Conjugation	$x^*(t)$	$X^*(-j\omega)$
4.3.5	Time Reversal	$x(-t)$	$X(-j\omega)$
4.3.5	Time and Frequency Scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(\frac{j\omega}{a}\right)$
4.4	Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
4.5	Multiplication	$x(t)y(t)$	$\frac{1}{2\pi}\int_{-\infty}^{+\infty} X(j\theta)Y(j(\omega-\theta))d\theta$
4.3.4	Differentiation in Time	$\frac{d}{dt}x(t)$	$j\omega X(j\omega)$
4.3.4	Integration	$\int_{-\infty}^{t} x(t)dt$	$\frac{1}{j\omega}X(j\omega) + \pi X(0)\delta(\omega)$
4.3.6	Differentiation in Frequency	$tx(t)$	$j\frac{d}{d\omega}X(j\omega)$
4.3.3	Conjugate Symmetry for Real Signals	$x(t)$ real	$\begin{cases} X(j\omega) = X^*(-j\omega) \\ \mathcal{R}e\{X(j\omega)\} = \mathcal{R}e\{X(-j\omega)\} \\ \mathcal{I}m\{X(j\omega)\} = -\mathcal{I}m\{X(-j\omega)\} \\ \lvert X(j\omega)\rvert = \lvert X(-j\omega)\rvert \\ \sphericalangle X(j\omega) = -\sphericalangle X(-j\omega) \end{cases}$
4.3.3	Symmetry for Real and Even Signals	$x(t)$ real and even	$X(j\omega)$ real and even
4.3.3	Symmetry for Real and Odd Signals	$x(t)$ real and odd	$X(j\omega)$ purely imaginary and odd
4.3.3	Even-Odd Decomposition for Real Signals	$x_e(t) = \mathcal{E}v\{x(t)\}$ [$x(t)$ real] $x_o(t) = \mathcal{O}d\{x(t)\}$ [$x(t)$ real]	$\mathcal{R}e\{X(j\omega)\}$ $j\mathcal{I}m\{X(j\omega)\}$
4.3.7	Parseval's Relation for Aperiodic Signals $\int_{-\infty}^{+\infty} \lvert x(t)\rvert^2 dt = \frac{1}{2\pi}\int_{-\infty}^{+\infty} \lvert X(j\omega)\rvert^2 d\omega$		

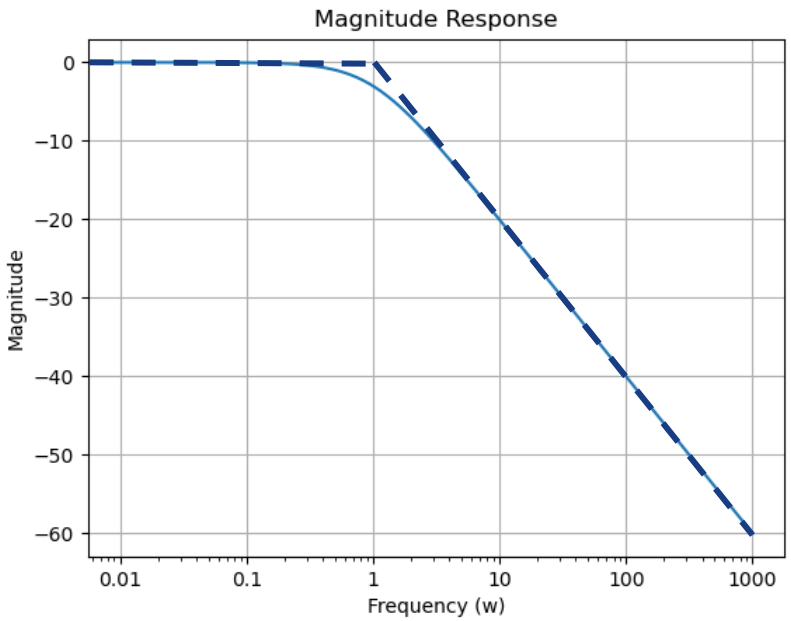
BODE PLOT

Bode plot is another way to visualize the frequency response

- **logarithmic frequency scale**
- **log magnitude, i.e. $20\,log_{10}|H(j\omega)|$ (dB)**

Example (Let $RC = 1$):

$$H(j\omega) = \frac{1}{1 + RCj\omega}$$



D.T. FOURIER SERIES FOR PERIODIC SIGNALS

D.T. Fourier series:

$$a_k = a_{k+N} = \frac{1}{N} \sum_{n=<N>} x[n]\, e^{-jk\Omega_0 n}$$

"Analysis" equation

$$x[n] = x[n+N] = \sum_{k=<N>} a_k\, e^{jk\Omega_0 n}$$

"Synthesis" equation

D.T. FOURIER SERIES PROPERTIES

TABLE 3.2 PROPERTIES OF DISCRETE-TIME FOURIER SERIES

Property	Periodic Signal	Fourier Series Coefficients
	$\left.\begin{matrix} x[n] \\ y[n] \end{matrix}\right\}$ Periodic with period N and fundamental frequency $\omega_0 = 2\pi/N$	$\left.\begin{matrix} a_k \\ b_k \end{matrix}\right\}$ Periodic with period N
Linearity	$Ax[n] + By[n]$	$Aa_k + Bb_k$
Time Shifting	$x[n - n_0]$	$a_k e^{-jk(2\pi/N)n_0}$
Frequency Shifting	$e^{jM(2\pi/N)n} x[n]$	a_{k-M}
Conjugation	$x^*[n]$	a^*_{-k}
Time Reversal	$x[-n]$	a_{-k}
Time Scaling	$x_{(m)}[n] = \begin{cases} x[n/m], & \text{if } n \text{ is a multiple of } m \\ 0, & \text{if } n \text{ is not a multiple of } m \end{cases}$ (periodic with period mN)	$\frac{1}{m} a_k \begin{pmatrix}\text{viewed as periodic} \\ \text{with period } mN\end{pmatrix}$
Periodic Convolution	$\sum_{r=\langle N\rangle} x[r]y[n-r]$	$Na_k b_k$
Multiplication	$x[n]y[n]$	$\sum_{l=\langle N\rangle} a_l b_{k-l}$
First Difference	$x[n] - x[n-1]$	$(1 - e^{-jk(2\pi/N)})a_k$
Running Sum	$\sum_{k=-\infty}^{n} x[k] \begin{pmatrix}\text{finite valued and periodic only} \\ \text{if } a_0 = 0\end{pmatrix}$	$\left(\frac{1}{(1 - e^{-jk(2\pi/N)})}\right) a_k$
Conjugate Symmetry for Real Signals	$x[n]$ real	$\begin{cases} a_k = a^*_{-k} \\ \mathcal{Re}\{a_k\} = \mathcal{Re}\{a_{-k}\} \\ \mathcal{Im}\{a_k\} = -\mathcal{Im}\{a_{-k}\} \\ \lvert a_k\rvert = \lvert a_{-k}\rvert \\ \sphericalangle a_k = -\sphericalangle a_{-k} \end{cases}$
Real and Even Signals	$x[n]$ real and even	a_k real and even
Real and Odd Signals	$x[n]$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals	$\begin{cases} x_e[n] = \mathcal{Ev}\{x[n]\} & [x[n] \text{ real}] \\ x_o[n] = \mathcal{Od}\{x[n]\} & [x[n] \text{ real}] \end{cases}$	$\mathcal{Re}\{a_k\}$ $j\mathcal{Im}\{a_k\}$

Parseval's Relation for Periodic Signals

$$\frac{1}{N} \sum_{n=\langle N\rangle} |x[n]|^2 = \sum_{k=\langle N\rangle} |a_k|^2$$

D.T. FOURIER TRANSFORM FOR APERIODIC SIGNALS

D.T. Fourier transform relations:

- Analysis equation (D.T. Fourier transform):

$$X\left(e^{j\Omega}\right) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\Omega n}$$

- Synthesis equation (D.T. inverse Fourier transform):

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\Omega})e^{j\Omega n}d\Omega$$

D.T. FOURIER TRANSFORM OF PERIODIC SIGNALS

Fourier transform representations for periodic signals allow us to consider both periodic and aperiodic signals within a unified context.

- More generally, if $x[n]$ is linear combination of complex exponentials equally spaced in frequency:

$$x[n] = \sum_{k=<N>} a_k \, e^{jk\Omega_0 n}$$

- The Fourier transform can be directly constructed from its Fourier coefficients!

$$X\left(e^{j\Omega}\right) = \sum_{k=-\infty}^{\infty} 2\pi a_k \, \delta(\Omega - k\Omega_0)$$

D.T. FOURIER TRANSFORM PROPERTIES

TABLE 5.1 PROPERTIES OF THE DISCRETE-TIME FOURIER TRANSFORM

Section	Property	Aperiodic Signal	Fourier Transform
		$x[n]$	$X(e^{j\omega})$ periodic with
		$y[n]$	$Y(e^{j\omega})$ period 2π
5.3.2	Linearity	$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$
5.3.3	Time Shifting	$x[n - n_0]$	$e^{-j\omega n_0}X(e^{j\omega})$
5.3.3	Frequency Shifting	$e^{j\omega_0 n}x[n]$	$X(e^{j(\omega-\omega_0)})$
5.3.4	Conjugation	$x^*[n]$	$X^*(e^{-j\omega})$
5.3.6	Time Reversal	$x[-n]$	$X(e^{-j\omega})$
5.3.7	Time Expansion	$x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n = \text{multiple of } k \\ 0, & \text{if } n \neq \text{multiple of } k \end{cases}$	$X(e^{jk\omega})$
5.4	Convolution	$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$
5.5	Multiplication	$x[n]y[n]$	$\frac{1}{2\pi}\int_{2\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$
5.3.5	Differencing in Time	$x[n] - x[n-1]$	$(1 - e^{-j\omega})X(e^{j\omega})$
5.3.5	Accumulation	$\sum_{k=-\infty}^{n} x[k]$	$\frac{1}{1 - e^{-j\omega}}X(e^{j\omega})$ $+ \pi X(e^{j0}) \sum_{k=-\infty}^{+\infty} \delta(\omega - 2\pi k)$
5.3.8	Differentiation in Frequency	$nx[n]$	$j\frac{dX(e^{j\omega})}{d\omega}$
5.3.4	Conjugate Symmetry for Real Signals	$x[n]$ real	$\begin{cases} X(e^{j\omega}) = X^*(e^{-j\omega}) \\ \mathcal{Re}\{X(e^{j\omega})\} = \mathcal{Re}\{X(e^{-j\omega})\} \\ \mathcal{Im}\{X(e^{j\omega})\} = -\mathcal{Im}\{X(e^{-j\omega})\} \\ \lvert X(e^{j\omega})\rvert = \lvert X(e^{-j\omega})\rvert \\ \sphericalangle X(e^{j\omega}) = -\sphericalangle X(e^{-j\omega}) \end{cases}$
5.3.4	Symmetry for Real, Even Signals	$x[n]$ real an even	$X(e^{j\omega})$ real and even
5.3.4	Symmetry for Real, Odd Signals	$x[n]$ real and odd	$X(e^{j\omega})$ purely imaginary and odd
5.3.4	Even-odd Decomposition of Real Signals	$x_e[n] = \mathcal{Ev}\{x[n]\}$ $[x[n]$ real$]$	$\mathcal{Re}\{X(e^{j\omega})\}$
		$x_o[n] = \mathcal{Od}\{x[n]\}$ $[x[n]$ real$]$	$j\mathcal{Im}\{X(e^{j\omega})\}$
5.3.9	Parseval's Relation for Aperiodic Signals $\sum_{n=-\infty}^{+\infty} \lvert x[n]\rvert^2 = \frac{1}{2\pi}\int_{2\pi} \lvert X(e^{j\omega})\rvert^2 d\omega$		

THE SAMPLING THEOREM

If signal is bandlimited → sample without loosing information.

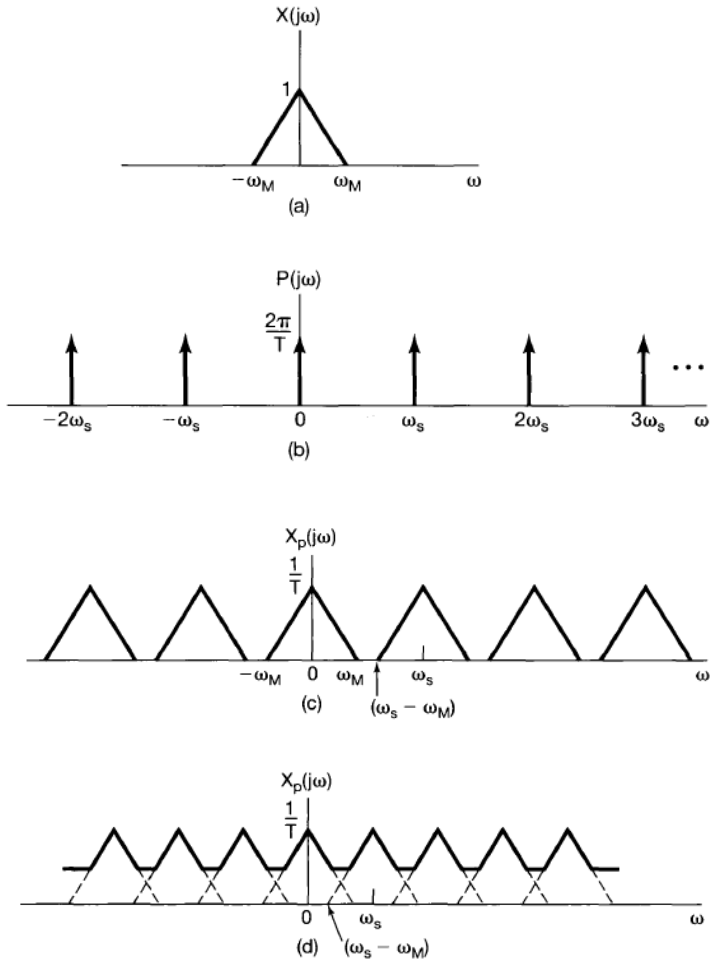
If $x(t)$ is bandlimited so that

$\mathrm{X}(j\omega) = 0$ for $|\omega| > \omega_m$

then $x(t)$ is uniquely determined by its samples $x(nT)$ if

$\omega_s - \omega_m > \omega_m \Longrightarrow \boldsymbol{\omega_s > 2\omega_m}$

The minimum sampling frequency, $2\omega_m$, is called the "Nyquist rate."



LAPLACE TRANSFORM PAIR

"Analysis" equation :

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt$$

Laplace transform

"Synthesis" equation:

$$x(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} X(s)\, e^{st}ds$$

Inverse Laplace transform

RELATION BETWEEN FOURIER AND LAPLACE TRANSFORM

Laplace transform and Fourier transform properties are related.

Property	$x(t)$	$X(j\omega)$	$X(s)$
Linearity	$ax_1(t) + bx_2(t)$	$aX_1(j\omega) + bX_2(j\omega)$	$aX_1(s) + bX_2(s)$
Time shift	$x(t - t_0)$	$e^{-j\omega t_0}X(j\omega)$	$e^{-st_0}X(s)$
Time scale	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(\frac{j\omega}{a}\right)$	$\frac{1}{\lvert a\rvert}X\left(\frac{s}{a}\right)$
Differentiation	$\frac{dx(t)}{dt}$	$j\omega X(j\omega)$	$sX(s)$
Multiply by t	$tx(t)$	$-\frac{1}{j}\frac{d}{d\omega}X(j\omega)$	$-\frac{d}{ds}X(s)$
Convolution	$x_1(t) * x_2(t)$	$X_1(j\omega) \times X_2(j\omega)$	$X_1(s) \times X_2(s)$

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